

Class 10 NCERT Maths

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Chapter 7: Coordinate Geometry (Core-Basics Edition)

CHAPTER 7: COORDINATE GEOMETRY

A simplified, detailed guide in Hinglish

Special edition for students who skipped Class 7-8-9

IS PDF MEIN RANG KA MATLAB:

RED text = core basic / prerequisite (Class 7-8-9 ka).

GREEN text = exam mein BAAR-BAAR aata hai (zaroor yaad rakh).

(Aur black = normal explanation.)

CORE BASICS - MISS MAT KARNA (RED PAGE)

1) X-AXIS, Y-AXIS, ORIGIN

X-axis = leti (horizontal) line. Y-axis = khadi line.
Dono milte hain ORIGIN (0, 0) pe.

2) COORDINATES (x, y)

Point ko (x, y) likhte hain. x = abscissa (X pe),
y = ordinate (Y pe). Order important: (3,5) $\neq$ (5,3).

3) SIGNS / QUADRANTS

Q1 (+,+), Q2 (-,+), Q3 (-,-), Q4 (+,-).
X-axis pe $y=0$, Y-axis pe $x=0$.

4) SQUARE AUR SQUARE ROOT

$a^2 = a \times a$. $\sqrt{25} = 5$.
Distance formula mein squares aur sqrt use hote hain.

5) NEGATIVE KA SQUARE

$(-4)^2 = 16$ (negative ka square POSITIVE hota hai).

6) AVERAGE (mean)

Do numbers ka average = $(a + b)/2$.
Midpoint formula mein bas average lena hai.

7) RATIO ($m_1 : m_2$)

Section formula mein point segment ko $m_1:m_2$ mein
baant-ta hai. Ratio = do hisson ki tulna.

NOW: ASLI CHAPTER 7 SHURU

TOPIC 1: DISTANCE FORMULA

EXAM ALERT: Distance formula har exam mein use hota:

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

(Ye Pythagoras se banaa hai.)

Origin se distance: $\sqrt{x^2 + y^2}$.

[CORE] differences ko square karte hain, isliye sign automatically positive ho jaata (RED page point 5).

TOPIC 2: SECTION FORMULA

EXAM ALERT: Point jo A,B ko $m_1:m_2$ mein baant-ta hai:

$$P = \left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$$

TOPIC 3: MIDPOINT FORMULA

EXAM ALERT: Midpoint (1:1 ka special case):

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

(Bas dono x ka average, dono y ka average.)

NOTE: Latest NCERT (2023-24) mein "Area of Triangle" wala part hata diya gaya hai, isliye include nahi kiya.

SOLVED EXAMPLES: HARDEST -> EASIEST

8 examples poori tarah solve karke. Upar HARDEST, neeche EASIEST.

Solved Example 1 (HARDEST) - Type of triangle

(Exam favourite - distance ka use)

A(3,4), B(-4,3), O(0,0). Dikhao OA = OB (isosceles).

$$OA = \sqrt{3^2 + 4^2} = \sqrt{9+16} = \sqrt{25} = 5$$

$$OB = \sqrt{(-4)^2 + 3^2} = \sqrt{16+9} = \sqrt{25} = 5$$

OA = OB = 5 -> triangle OAB isosceles hai.

Solved Example 2 - Collinear points

(Exam favourite)

Kya A(1,5), B(2,3), C(-2,-11) collinear hain?

(Collinear = AB + BC = AC ho.)

$$AB = \sqrt{(2-1)^2 + (3-5)^2} = \sqrt{1+4} = \sqrt{5} = 2.24$$

$$BC = \sqrt{(-2-2)^2 + (-11-3)^2} = \sqrt{16+196} = \sqrt{212} = 14.56$$

$$AC = \sqrt{(-2-1)^2 + (-11-5)^2} = \sqrt{9+256} = \sqrt{265} = 16.28$$

AB + BC (16.8) != AC (16.28) -> NOT collinear.

Solved Example 3 - Section formula

(Exam favourite)

A(4,-1), B(-2,3) ko 2:1 mein baant-ne wala point.

$$m_1=2, m_2=1.$$

$$x = \frac{2(-2) + 1(4)}{3} = \frac{-4+4}{3} = 0$$

$$y = \frac{2(3) + 1(-1)}{3} = \frac{6-1}{3} = \frac{5}{3}$$

Point = (0, 5/3).

Solved Example 4 - Find missing end (midpoint)

Midpoint (3,4) hai, ek end A(1,2). Doosra end B?

$$\frac{1+x}{2} = 3 \rightarrow x = 5$$

$$\frac{2+y}{2} = 4 \rightarrow y = 6$$

B = (5, 6).

Solved Example 5 - Distance between 2 points

A(2,3), B(5,7). Distance?

$$AB = \sqrt{(5-2)^2 + (7-3)^2} = \sqrt{9+16} = \sqrt{25} = 5.$$

Solved Example 6 - Midpoint

A(4,6), B(8,10). Midpoint?

$$M = ((4+8)/2, (6+10)/2) = (6, 8).$$

Solved Example 7 - Distance from origin

P(6,8) ka origin se distance?

$$OP = \sqrt{6^2 + 8^2} = \sqrt{36+64} = \sqrt{100} = 10.$$

Solved Example 8 (EASIEST) - Quadrant batao

(5,-3) kis quadrant mein? -> x +ve, y -ve -> Q4.

(-2,-7) -> dono -ve -> Q3.

Q AND A TIME

Q1. CORE: in points ka quadrant: (4,8), (-1,6), (3,-2).

Q2. A(0,0), B(6,8) ke beech distance.

Q3. P(3,4) ka origin se distance.

Q4. A(-1,2), B(5,8) ka midpoint.

Q5. A(1,2), B(7,8) ko 1:2 mein baant-ne wala point.

Q6. Kya A(0,0), B(3,4) ke beech distance 5 hai? (Check)

Q7. CORE CHECK: $(-3)^2$ kya hota hai?

SUMMARY

1. Point = (x, y); x abscissa, y ordinate.
2. Distance: $\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$.
3. Section (m1:m2): $\left(\frac{m_1x_2+m_2x_1}{m_1+m_2}, \frac{m_1y_2+m_2y_1}{m_1+m_2}\right)$.
4. Midpoint: $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$.
5. Quadrants: Q1(+,+), Q2(-,+), Q3(-,-), Q4(+,-).

CORE (RED page) revise: axes/origin, coordinates, signs, square aur sqrt, average, ratio.